Memory Production under Coarse-Graining: A Balance Law for the Indivisibility Deficit*

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Abstract

Eliminating variables from a Markov process generically produces memory: the reduced two-time transition family violates the Chapman–Kolmogorov property although the full family satisfies it. This paper quantifies that production with the liftability profile introduced for transition families without a generator. In the time-inhomogeneous dilation class, $h_d(V)$ is the attained distance from V to the compressions of d-state flat transports, and at the dimension-matched value $d = n$ the profile is the distance from V to the flat families, the charge of the family. The charge is not monotone under coarse-graining of the state space: a lumped Markov chain has a reduced family with positive charge although the fine family has none, and a product family with positive charge has a flat reduction when its memory-bearing factor is discarded. The deficit obeys an exact balance law instead. For a lumping of a family with strictly positive laws, the fine deficit splits orthogonally into a flux term (its compression through the cut, which is contractive in the induced metric) and a severed remainder invisible to observables measurable with respect to the coarse partition; the coarse deficit is the flux plus a production term sourced at the cut. Production vanishes exactly when the cut is closed on the accessible subspace. For a flat fine family the zero set of production is two-time Chapman–Kolmogorov lumpability, composition of the observed two-time family; weak lumpability implies it, and the converse fails. The charge inherits two-sided control from the deficit norm with explicit constants and exponent one, so the balance law transfers to the charge as bounds. For a flat fine family the production term is the two-time Mori–Zwanzig memory kernel of the law-induced conditional-expectation projection, and it equals the deficit of the observed family, which is defined without a projection; among block-supported stochastic sections the law section is the unique one whose kernel equals the deficit. Along a tower of coarse-grainings in which each cut severs nothing and each production term is coherent with the transmitted deficit, the deficit norm does not decrease. Both hypotheses are sharp: discarding a memory-bearing autonomous factor severs its deficit in full, and an explicit perfect-interference construction with a strictly positive joint law severs nothing, has production equal to the negative of the transmitted deficit, and has a flat coarse family although the lumped process is not Markov. Every displayed quantity is an exact rational number or an algebraic number in closed form.

Keywords. Chapman–Kolmogorov divisibility, lumpability, coarse-graining, memory kernel, Mori–Zwanzig formalism, indivisible stochastic processes, balance law

AMS subject classifications. 60J10, 82C31, 60G05, 15B51

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Significance statement

A reduced description of a dynamical system is obtained by a cut: some variables are kept and the rest are eliminated. Projection-operator theory shows that the eliminated variables return as a memory kernel, and the theory of lumped Markov chains shows that a lumped chain generically fails the Chapman–Kolmogorov property. This paper connects the two facts by an exact balance law for the two-time Chapman–Kolmogorov deficit; the charge of the companion projection theory is a distance-based measure of this deficit. Across any cut the fine deficit splits orthogonally into a transmitted part and a severed part, and the coarse deficit is the transmitted part plus a production term sourced at the cut. The two-time composition defect is produced at cuts that are not closed on the accessible subspace, destroyed at cuts that sever the directions carrying it, cancelled at cuts whose production is opposite to the transmitted part, and conserved when none of these happens. For flat fine dynamics the production term is the two-time Mori–Zwanzig memory kernel of the law-induced projection and its zero set is two-time Chapman–Kolmogorov lumpability. The balance law concerns the two-time deficit and says nothing about the higher-order dependence of the lumped process.

1 Introduction

Let $X = (X_{t_i})_{i=0}^{m}$ be a stochastic process on a finite state space $[N]$, observed on a finite time grid, and let V be its two-time transition family, $V(j \leftarrow i)_{vu} = \mathbb{P}(X_{t_j} = v \mid X_{t_i} = u)$. If X is Markov, the family composes,

$$V(t \leftarrow s) \;=\; V(t \leftarrow r)\, V(r \leftarrow s), \qquad s \le r \le t, \tag{1}$$

the Chapman–Kolmogorov property. Fix a surjection $\lambda : [N] \to [n]$ with $n < N$ and observe $Y = \lambda(X)$. The transition family of Y generically violates (1) although the family of X satisfies it [8, 2]. The violation is the deficit

$$\Delta_V(t, s, r) \;=\; V(t \leftarrow s) - V(t \leftarrow r)\, V(r \leftarrow s), \qquad s < r < t, \tag{2}$$

the central object of the program of which this paper is a part [18, 19, 20, 21, 22], and the two-time counterpart of the memory kernel that projection-operator techniques attach to a reduced description [12, 27, 11, 7, 28, 24]. The companion paper [22] converts the deficit into an invariant. In the time-inhomogeneous dilation class of [22], the liftability profile $h_d(V)$ is the attained distance from V to the compact model set $\mathcal{F}_d$ of compressions of d-state flat transports, and at $d = n$ the model set is the set of flat families [22, Example 5.2], so that

$$h_n(V) \;=\; \operatorname{dist}(V, \mathcal{F}\mathrm{lat}_n). \tag{3}$$

We call $h_n(V)$ the *charge* of the family.[1] It is a distance-based measure of the Chapman–Kolmogorov failure; Proposition 2.6 relates it to the metric size $\|\Delta_V\|$ of the failure with explicit constants. In the time-homogeneous class of [22] the model set at $d = n$ is strictly smaller than $\mathcal{F}\mathrm{lat}_n$ and (3) fails [22, Section 7]; this paper works in the time-inhomogeneous class throughout.

This paper asks how the deficit, and with it the charge, transforms under coarse-graining of the state space. The charge is monotone in neither direction. Elimination of variables creates memory, which suggests that the charge grows as the description coarsens; coarse-graining destroys

[1] Numbered references to [22] are to its September 2026 revision; every cited number is unchanged from the SSRN preprint.

information, which suggests that it shrinks. Section 3 gives exact counterexamples to both: a three-state Markov chain whose two-block lumping has a coarse deficit of metric size $\sqrt{18/23575}$ although the fine family is flat, and a four-state product family with positive charge whose coarse-graining onto its memoryless factor is flat.

Section 4 proves the balance law that holds instead. For a lumping of a family with strictly positive laws, the coarse deficit decomposes as

$$\Delta_{V^\lambda}(t,s,r) \;=\; \underbrace{\mathcal{C}_s\big[\Delta_V(t,s,r)\big]}_{\text{flux}} \;+\; \underbrace{\mathrm{P}_\lambda(t,s,r)}_{\text{production}}, \tag{4}$$

where the flux term is the compression of the fine deficit through the cut and the production term is sourced at the cut. The compression $\mathcal{C}$ is a conditional expectation in the metric weighted by the laws of the process (Lemma 2.3), so it is contractive, and the fine deficit splits orthogonally into a transmitted component and a *severed* component,

$$\|\Delta_V\|^2 \;=\; \|\mathcal{C}[\Delta_V]\|^2 + \|\Xi_\lambda\|^2. \tag{5}$$

The identity (4) is algebraic and holds in every metric. The orthogonality in (5), the contraction of the flux, and the sizes of the terms are properties of the law-weighted metric (6), and the zero sets of all quantities are independent of the metric (Section 2). Production vanishes exactly when the cut is closed on the accessible subspace, that is, when the conditional law of the coarse future given the fine present is constant on the blocks along the directions the process reaches from earlier times (Theorem 4.5). For a flat fine family the zero set of production is two-time Chapman–Kolmogorov lumpability: the observed two-time family composes. Weak lumpability in the sense of Kemeny and Snell, Markovianity of the lumped process for the given initial law, implies it, and strong lumpability implies it; neither implication reverses (Corollary 4.6, Example 4.7), and the lumped process of the interference example of Section 6 has a flat coarse family and is not Markov (Remark 6.8).

Section 5 identifies the production term. With Π_r the conditional-expectation projection onto coarse observables at time t_r and $Q_r = I - \Pi_r$,

$$\mathrm{P}_\lambda(t,s,r) \;=\; L\,V(t \leftarrow r)\,Q_r\,V(r \leftarrow s)\,R_s :$$

propagation from s to the cut time r, projection onto the component invisible to the coarse description, propagation to t, observation. For a flat fine family this is the two-time Mori–Zwanzig memory kernel of the law-induced projection, and the discrete Dyson expansion of Proposition 5.1 writes the coarse family as a sum over resolved and fluctuation segments. The formalism ordinarily depends on the choice of projection. Theorem 5.2 states what is independent of that choice: the deficit of the observed family is defined without a projection; for a flat fine family the kernel of the law-induced projection equals it; and among block-supported stochastic sections the law section is the unique one with that property. The kernels of Mori's linear projections and of Wiener projections are not of that form and are not covered by the uniqueness statement. The charge measures the production term two-sidedly with the explicit constants of Proposition 2.6; the residual at a nearest flat family has norm h_n and is not unique in general (Example 5.4, a family with laws that is not process-induced).

Section 6 treats towers of coarse-grainings. Along a tower step that severs nothing and whose production is coherent with the transmitted deficit, the deficit norm does not decrease, and it increases strictly when production is present (Theorem 6.2). Both hypotheses are sharp. Projecting away an independent factor is a dynamically closed cut whose flux is the retained factor's deficit,

so the discarded factor's deficit is severed in full (Proposition 6.4). Proposition 6.5 constructs a three-state process with a displayed strictly positive joint law whose two-block cut severs nothing and has production equal to the negative of the transmitted deficit, so that the coarse family is flat; the lumped process is not Markov (Remark 6.8), so the cut cancels the two-time composition defect and not the dependence of the lumped process on its past. Section 7 lists the four families of the paper with every entry an exact rational number and the charges as exact windows, Section 8 states the conclusion and the limits of the results, and Section 9 states open problems.

Positioning. The orthogonal splitting (5) has an antecedent in information geometry. Wolfer and Watanabe [26] establish a Pythagorean decomposition $D(P\|P') = D(P\|P_0) + D(P_0\|P')$ for the data-processing of Markov chains relative to a projection onto the lumpable family. The geometric content, an orthogonal decomposition relative to a projection onto coarse structure with contraction on one side, is the same; the geometry there is the dually flat Kullback–Leibler geometry of chains; the geometry here is a law-weighted quadratic geometry of deficit cochains; and the production term in (4) has no counterpart there. In open quantum systems the failure of divisibility of a reduced dynamical map is a measure of non-Markovianity [14, 10], and information backflow [1] is a strictly stronger property: divisibility implies the absence of backflow, and the converse fails [4]. Production sources the failure of two-time divisibility; its comparison with backflow is an analogy. The composition test of Corollary 4.6 has an empirical counterpart in the Chapman–Kolmogorov test of Markov-state-model validation [23, 13], which tests an estimated coarse family for composition at multiples of the lag time; Open Problem (iv) is posed with reference to it.

The contributions, in the order of the paper, are: (i) exact counterexamples to monotonicity of the charge in both directions (Theorem 3.1); (ii) the compression-as-conditional-expectation lemma and the explicit two-sided comparison between the charge and the deficit norm, with exponent one (Lemma 2.3, Proposition 2.6); (iii) the exact balance law with the conservation identity for the severed component (Theorem 4.2); (iv) the characterization of zero production by closure on the accessible subspace, and two-time Chapman–Kolmogorov lumpability as its zero set for flat fine families (Theorem 4.5, Corollary 4.6); (v) the identification of production as the two-time Mori–Zwanzig kernel of the law-induced projection, the uniqueness statement among admissible sections, and the non-uniqueness of the residual at a nearest flat family, shown outside the process-induced class (Section 5); (vi) monotonicity of the deficit norm along towers with sharp hypotheses, the autonomy proposition, and the perfect-interference construction (Section 6); (vii) exact examples for the four extremal mechanisms of the balance law (Section 7). The paper contains no numerical optimization, no simulation, and no data; every displayed quantity is an exact rational number or an algebraic number given in closed form.

2 Setting: families, the law metric, and cuts

2.1 Families, the metric, and the charge

Notation follows [22]. Fix a finite time grid $\mathbb{T} = \{t_0 < \cdots < t_m\}$, $m \geq 2$, written $0, \ldots, m$. For an integer $k \geq 2$ let $\mathcal{S}_k$ be the column-stochastic $k \times k$ matrices, and let $\mathcal{V}_k = \prod_{i<j} \mathcal{S}_k$ be the transition families on k states, with elements $V = \{V(j \leftarrow i)\}_{i<j}$. A family is *flat* if it satisfies (1) on the grid; the set of flat families is $\mathcal{F}\mathrm{lat}_k \subset \mathcal{V}_k$, the zero set of the deficit (2). A family is *process-induced* if it is the two-time conditional family of a process on $[k]$ with strictly positive one-time laws.

Throughout, a family carries *laws*: strictly positive probability vectors $p_0, \ldots, p_m$ on $[k]$, supplied by the process when the family is process-induced, in which case $V(j \leftarrow i)\, p_i = p_j$ for all

$i < j$. The laws enter the metric. For kernel differences A, B attached to the pair $(i < j)$ define

$$\langle A, B\rangle_{(j\leftarrow i)} \;=\; \sum_{v,u=1}^{k} \frac{p_i(u)}{p_j(v)}\, A_{vu} B_{vu}, \qquad \|A\|_{(j\leftarrow i)} = \langle A, A\rangle_{(j\leftarrow i)}^{1/2}, \tag{6}$$

and for families the sum over pairs with unit pair weights; deficits $\Delta(t, s, r)$ carry the $(t \leftarrow s)$ weights, and deficit cochains the sum over triples.

The metric (6) is not in the admissible class of [22]: the weights of that class depend on the source state only, and (6) depends on the destination state through $1/p_j(v)$. It is equivalent to every member of that class. With the class weight $p_i(u)$ at the pair $(i < j)$,

$$\big(\max_v p_j(v)\big)^{-1/2} \|A\|_{[\mathrm{D}]} \;\le\; \|A\|_{(j\leftarrow i)} \;\le\; \big(\min_v p_j(v)\big)^{-1/2} \|A\|_{[\mathrm{D}]},$$

so the zero sets of all distances in this paper coincide with those of [22] (the argument of [22, Lemma 4.8]: equivalent norms have the same zero-distance sets), and the comparison [22, Proposition 5.3] holds in (6) with constants adjusted by the equivalence constants. Proposition 2.6 proves that comparison at $d = n$ directly in (6), with explicit constants and exponent one. The metric (6) is the choice under which compression through a cut is a conditional expectation (Lemma 2.3). The identities of this paper that equate kernels hold in every metric; the statements about orthogonality, contraction, and the sizes of terms are statements about (6).

The *charge* of a family is its dimension-matched liftability value (3), $h_n(V) = \mathrm{dist}(V, \mathcal{F}\mathrm{lat}_n)$ in the metric (6), taken in the time-inhomogeneous dilation class of [22], in which the dimension-matched model set is $\mathcal{F}\mathrm{lat}_n$ [22, Example 5.2]. Since $\mathcal{F}\mathrm{lat}_n$ is the zero set of the deficit, $h_n(V) = 0$ if and only if $\Delta_V = 0$; the quantitative relation between $h_n(V)$ and $\|\Delta_V\|$ is Proposition 2.6. The balance law of this paper is exact at the level of the deficit cochain, and it transfers to the charge as two-sided bounds through that proposition.

2.2 Cuts: lumpings, compression, disintegration

Definition 2.1 (Cut)**.** A *cut* of $[N]$ is a surjection $\lambda : [N] \to [n]$, $2 \le n < N$, encoded by the 0/1 matrix $L \in \mathbb{R}^{n\times N}$, $L_{au} = \mathbf{1}\{\lambda(u) = a\}$, so that $\mathbf{1}^\top L = \mathbf{1}^\top$. Since the laws p_i are strictly positive, the block masses $p_i^\lambda = L p_i$ are strictly positive. The *disintegration* of p_i over the blocks is the column-stochastic section $R_i \in \mathbb{R}^{N\times n}$,

$$(R_i)_{ua} \;=\; \frac{p_i(u)\,\mathbf{1}\{\lambda(u) = a\}}{p_i^\lambda(a)}, \qquad L R_i = I_n,$$

and $\Pi_i = R_i L \in \mathbb{R}^{N\times N}$ is the induced idempotent, $\Pi_i^2 = \Pi_i$, with fluctuation complement $Q_i = I - \Pi_i$. The *compression* of the family is

$$V^\lambda(j \leftarrow i) \;=\; L\, V(j \leftarrow i)\, R_i,$$

with laws p_i^λ; the compression of a kernel difference A at the pair $(i < j)$ is $\mathcal{C}_i[A] = LAR_i$.

Proposition 2.2 (The compression is the observed family)**.** *If V is the conditional family of a process X with laws p, then V^λ is the conditional family of the lumped process $Y = \lambda(X)$:*

$$V^\lambda(j \leftarrow i)_{ba} \;=\; \mathbb{P}\big(\lambda(X_{t_j}) = b \mid \lambda(X_{t_i}) = a\big),$$

with laws p_i^λ. In particular V^λ is process-induced and marginal-consistent, and its charge is the charge of the process observed through the cut.

Proof. $\big(LV(j \leftarrow i)R_i\big)_{ba} = \sum_{v:\lambda v=b} \sum_u V(j \leftarrow i)_{vu} \frac{p_i(u)\mathbf{1}\{\lambda u=a\}}{p_i^\lambda(a)} = \frac{\mathbb{P}(\lambda X_{t_j}=b,\ \lambda X_{t_i}=a)}{\mathbb{P}(\lambda X_{t_i}=a)}$ by the law of total probability. Marginal consistency $V^\lambda(j \leftarrow i)p_i^\lambda = p_j^\lambda$ follows from $V(j \leftarrow i)p_i = p_j$ and $R_i p_i^\lambda = \Pi_i p_i = p_i$. □

This is the classical setting in which the Chapman–Kolmogorov property fails [8, 2], and the compression coincides with the observable compression κ of the dilation theory of [22]: the charge measures the distance from a coarse family to $\mathcal{F}\mathrm{lat}_n$.

2.3 Compression as conditional expectation

Fix a pair $(i < j)$. Let $H = L^2([N]\times[N],\, p_j \otimes p_i)$, let $\mathcal{B}_\lambda$ be the sub-σ-algebra of $[N]\times[N]$ generated by $\lambda\times\lambda$, let $H_\lambda \subseteq H$ be the subspace of $\mathcal{B}_\lambda$-measurable functions, and let $H^\lambda = L^2([n]\times[n],\, p_j^\lambda \otimes p_i^\lambda)$. A kernel difference A at $(i < j)$ is identified with its density $f_A(v,u) = A_{vu}/p_j(v) \in H$, and a coarse kernel difference δ with $f_\delta(b,a) = \delta_{ba}/p_j^\lambda(b) \in H^\lambda$; both identifications are isometries for (6) at the respective level, since $\sum_{v,u} p_j(v)p_i(u)A_{vu}^2/p_j(v)^2 = \|A\|^2_{(j\leftarrow i)}$. The pull-back $f \mapsto f \circ (\lambda \times \lambda)$ is an isometry of H^λ onto H_λ. The compression $\mathcal{C}_i$ is not an operator on a single space; the lemma identifies it with the composition $H \to H_\lambda \to H^\lambda$ of the conditional expectation with the inverse pull-back.

Lemma 2.3 (Conditional expectation, isometric lift, Pythagoras)**.** *With the identifications above:*

(a) the density of the compression $\mathcal{C}_i[A] = LAR_i$ with respect to the coarse laws is the conditional expectation of f_A: $f_{\mathcal{C}_i[A]} \circ (\lambda \times \lambda) = \mathbb{E}_{p_j\otimes p_i}[f_A \,|\, \mathcal{B}_\lambda]$;

(b) the block lift $\iota_i[\delta]_{vu} = \delta_{\lambda v,\lambda u}\, p_j(v)/p_j^\lambda(\lambda v)$ is the kernel difference whose density is $f_\delta \circ (\lambda\times\lambda)$; it is an isometric right inverse of $\mathcal{C}_i$ with image the $\mathcal{B}_\lambda$-measurable kernel differences, and it preserves vanishing column sums;

(c) $\mathcal{C}_i$ maps H onto H^λ with $\|\mathcal{C}_i[A]\| \le \|A\|$ (contraction in the induced metric*, the metric* (6) *built from the lumped laws), and*

$$\|A\|^2 \;=\; \|\mathcal{C}_i[A]\|^2 \;+\; \|A - \iota_i\mathcal{C}_i[A]\|^2, \tag{7}$$

with $A - \iota_i\mathcal{C}_i[A]$ orthogonal to every block lift.

Proof. (a) For $b = \lambda v$, $a = \lambda u$,

$$\mathbb{E}[f_A|\mathcal{B}_\lambda]\,(v,u) = \frac{\sum_{v'\in\lambda^{-1}b}\sum_{u'\in\lambda^{-1}a} p_j(v')p_i(u')\,A_{v'u'}/p_j(v')}{p_j^\lambda(b)\,p_i^\lambda(a)} = \frac{\big(L\,A\,\mathrm{diag}(p_i)\,L^\top\big)_{ba}}{p_j^\lambda(b)\,p_i^\lambda(a)} = \frac{(LAR_i)_{ba}}{p_j^\lambda(b)},$$

using that $\big(LA\,\mathrm{diag}(p_i)L^\top\big)_{ba} = \sum_{v'\in\lambda^{-1}b}\sum_{u'\in\lambda^{-1}a} A_{v'u'}\,p_i(u')$ and $R_i = \mathrm{diag}(p_i)L^\top\mathrm{diag}(p_i^\lambda)^{-1}$. The right-hand side is the density of LAR_i with respect to $(p_j^\lambda, p_i^\lambda)$ pulled back along $\lambda\times\lambda$. (b) The density of $\iota_i[\delta]$ is $f_\delta\circ(\lambda\times\lambda)$ by direct substitution; $\mathcal{C}_i\iota_i = \mathrm{id}$ by direct computation; the isometry is that of the pull-back; and $\sum_v \iota_i[\delta]_{vu} = \sum_b \delta_{b,\lambda u} = 0$ when δ has vanishing column sums. (c) Conditional expectation is the orthogonal projection of H onto H_λ [25]; (a) and (b) transport orthogonality, contraction, and (7) to kernel differences. □

Remark 2.4. Lemma 2.3(b) states that coarse kernel differences, measured with the lumped laws, embed isometrically into fine kernel differences as the block-measurable subspace; "contractive in the induced metric" in the abstract refers to this pair of metrics. The lemma fails for generic entry

weights, and it fails for the source-weighted class of [22]: the conditional-expectation structure is a property of (6), and it is available because the process supplies its laws. For a norm of that class at the pair $(i<j)$, $\|A\|^2=\sum_{v,u}\omega_i(u)A_{vu}^2$ with a source weight ω_i and no target weight, and with the lumped weight $L\omega_i$ at the coarse level, the lift of Lemma 2.3(b) has

$$\|\iota_i[\delta]\|^2 \;=\; \sum_{a,b}(L\omega_i)(a)\,\delta_{ba}^2\sum_{v\in\lambda^{-1}(b)}\frac{p_j(v)^2}{p_j^{\lambda}(b)^2},$$

and the inner sum is less than 1 for every block with two or more states, so the lift is not an isometry, and (7) fails for $A=\iota_i[\delta]$ whenever δ has a nonzero entry in a row whose block has two or more states. An instance with the laws of a process: on the grid $\{0,1,2\}$ let $p=(\frac12,\frac14,\frac14)^\top$ at all three times, let $\lambda(1)=\lambda(2)=a$, $\lambda(3)=b$, with coarse law $(\frac34,\frac14)$, let $\delta_1=\left(\begin{smallmatrix}1&-3\\-1&3\end{smallmatrix}\right)$, which has vanishing column sums and annihilates the coarse law, so that $\iota_0[\delta_1]$ has vanishing column sums and $\iota_0[\delta_1]\,p=0$, and let

$$V(1\leftarrow 0)=V(2\leftarrow 1)=p\mathbf{1}^\top,\qquad V(2\leftarrow 0)=p\mathbf{1}^\top+\tfrac18\,\iota_0[\delta_1]=\begin{pmatrix}7/12&7/12&1/4\\7/24&7/24&1/8\\1/8&1/8&5/8\end{pmatrix}.$$

The family is process-induced: the joint law $\pi(u,v,w)=p(u)\,p(v)\,V(2\leftarrow 0)_{wu}\ge\frac{1}{128}$ has the two-time margins $p(u)p(v)$, $p(v)p(w)$, and $p(u)V(2\leftarrow 0)_{wu}$. Its deficit $\Delta_V(2,0,1)=\frac18\iota_0[\delta_1]$ is a block lift, so $\Phi_\lambda=\frac18\delta_1$ and $\Xi_\lambda=0$ (Definition 4.1), and in (6) $\|\Delta_V\|^2=\|\Phi_\lambda\|^2=\frac14$. With the weights $p_i(u)$ at the source and no target weight, $\|\Delta_V\|^2=\frac{7}{96}$ and $\|\Phi_\lambda\|^2=\frac{9}{96}$, so the identity $\|\Delta_V\|^2=\|\Phi_\lambda\|^2+\|\Xi_\lambda\|^2$ of Theorem 4.2(ii) fails by $\frac{1}{48}$ in that norm.

2.4 Explicit comparison between the charge and the deficit

For a column-stochastic K regarded as a kernel from time a to time b, define the *law distortion*

$$\varkappa_{b\leftarrow a}(K)\;=\;\max_v\frac{(Kp_a)(v)}{p_b(v)}.$$

Since Kp_a and p_b are probability vectors, $1\le\varkappa_{b\leftarrow a}(K)\le 1/\min_v p_b(v)$, and $\varkappa_{b\leftarrow a}(K)=1$ when $Kp_a=p_b$.

Lemma 2.5 (Multiplication bounds)**.** *Let K be a kernel from time a to time b with laws as above.*

(a) For a kernel difference E at the pair $(i<a)$: $\|KE\|_{(b\leftarrow i)}\le\varkappa_{b\leftarrow a}(K)^{1/2}\,\|E\|_{(a\leftarrow i)}$.

(b) For a kernel difference D at the pair $(b<t)$: $\|DK\|_{(t\leftarrow a)}\le\varkappa_{b\leftarrow a}(K)^{1/2}\,\|D\|_{(t\leftarrow b)}$.

Proof. Write $q=Kp_a$. (a) If $q(v)=0$, then the v-th row of K vanishes and so does the v-th row of KE; put $R_{wv}=0$ for such v. For v with $q(v)>0$ the reverse kernel $R_{wv}=K_{vw}p_a(w)/q(v)$ is a probability vector in w, and $(KE)_{vu}=q(v)\sum_w R_{wv}E_{wu}/p_a(w)$. By Jensen's inequality, $(KE)_{vu}^2\le q(v)^2\sum_w R_{wv}E_{wu}^2/p_a(w)^2$, so

$$\begin{aligned}\|KE\|_{(b\leftarrow i)}^2\;&\le\;\sum_{w,u}\frac{p_i(u)E_{wu}^2}{p_a(w)^2}\sum_v\frac{q(v)^2R_{wv}}{p_b(v)}\;=\;\sum_{w,u}\frac{p_i(u)E_{wu}^2}{p_a(w)^2}\sum_v\frac{q(v)}{p_b(v)}K_{vw}p_a(w)\\&\le\;\varkappa_{b\leftarrow a}(K)\,\|E\|_{(a\leftarrow i)}^2,\end{aligned}$$

since $\sum_v K_{vw} = 1$. (b) By Jensen's inequality with the probability weights $K_{\cdot u}$, $(DK)_{vu}^2 \le \sum_w D_{vw}^2 K_{wu}$, so

$$\|DK\|_{(t\leftarrow a)}^2 \;\le\; \sum_{v,w} \frac{D_{vw}^2}{p_t(v)} \sum_u p_a(u) K_{wu} \;=\; \sum_{v,w} \frac{p_b(w)}{p_t(v)} D_{vw}^2 \, \frac{q(w)}{p_b(w)} \;\le\; \varkappa_{b\leftarrow a}(K)\, \|D\|_{(t\leftarrow b)}^2. \qquad \square$$

Proposition 2.6 (Explicit comparison at the dimension-matched level)**.** *Let $V \in \mathcal{V}_n$ carry strictly positive laws p on the grid $\{0,\dots,m\}$, and let $\rho = \min_{l,v} p_l(v)$. Then*

$$\frac{\|\Delta_V\|}{L_*(m,\rho)} \;\le\; h_n(V) \;\le\; C_*(m,\rho)\, \|\Delta_V\|, \tag{8}$$

where

$$L_*(m,\rho) = \Big((m-1)\big(1+\tfrac{2}{\rho}\big)\Big)^{1/2}, \qquad C_*(m,\rho) = \Big((m-1)\sum_{q=0}^{m-2} \rho^{-q}\Big)^{1/2}.$$

On the three-point grid $\{0,1,2\}$,

$$\frac{\|\Delta_V\|}{\big(1 + 1/\min_w p_1(w) + 1/\min_v p_2(v)\big)^{1/2}} \;\le\; h_n(V) \;\le\; \|\Delta_V\|. \tag{9}$$

Proof. Upper bound. Let $M_k = V(k \leftarrow k{-}1)$ and let W be the flat family with $W(j \leftarrow i) = M_j \cdots M_{i+1}$. Then $W(i{+}1 \leftarrow i) = V(i{+}1 \leftarrow i)$, and for $j \ge i+2$, since $V(j \leftarrow i) = \Delta_V(j,i,j{-}1) + M_j V(j{-}1 \leftarrow i)$ and $W(j \leftarrow i) = M_j W(j{-}1 \leftarrow i)$, induction on j gives

$$V(j \leftarrow i) - W(j \leftarrow i) \;=\; \sum_{k=i+2}^{j} M_j \cdots M_{k+1}\, \Delta_V(k,i,k{-}1).$$

By Lemma 2.5(a) applied $j-k$ times, each term has norm at most $\rho^{-(j-k)/2}\|\Delta_V(k,i,k{-}1)\|_{(k\leftarrow i)}$, so by the Cauchy–Schwarz inequality

$$\|V(j \leftarrow i) - W(j \leftarrow i)\|^2 \;\le\; \Big(\sum_{q=0}^{m-2} \rho^{-q}\Big) \sum_{k=i+2}^{j} \|\Delta_V(k,i,k{-}1)\|^2.$$

Summing over the pairs (i,j) with $j \ge i{+}2$ counts each triple $(k,i,k{-}1)$ once for each $j = k,\dots,m$, at most $m-1$ times, so $\operatorname{dist}(V,W)^2 \le C_*(m,\rho)^2 \|\Delta_V\|^2$, and $h_n(V) \le \operatorname{dist}(V,W)$. On the three-point grid the sum has one term with $q=0$ and $m-1=1$, so $C_*(2,\rho) = 1$.

Lower bound. Let W be any flat family. For $s < r < t$, since $\Delta_W = 0$,

$$\begin{aligned}\Delta_V(t,s,r) \;=\; &\big(V(t\leftarrow s) - W(t\leftarrow s)\big) - \big(V(t\leftarrow r) - W(t\leftarrow r)\big)V(r\leftarrow s) \\ &\qquad - W(t\leftarrow r)\big(V(r\leftarrow s) - W(r\leftarrow s)\big).\end{aligned}$$

By Lemma 2.5(b) with $K = V(r\leftarrow s)$ and (a) with $K = W(t\leftarrow r)$, and then the Cauchy–Schwarz inequality on the three terms,

$$\|\Delta_V(t,s,r)\|^2 \;\le\; \Big(1 + \frac{1}{\min_w p_r(w)} + \frac{1}{\min_v p_t(v)}\Big)\Big(\|V_{ts} - W_{ts}\|^2 + \|V_{tr} - W_{tr}\|^2 + \|V_{rs} - W_{rs}\|^2\Big),$$

with $V_{ts} = V(t \leftarrow s)$ and so on. Each pair of grid points lies in exactly $m-1$ triples, so summing over triples gives $\|\Delta_V\|^2 \le (1 + 2/\rho)(m-1)\operatorname{dist}(V,W)^2$. Taking W at distance $h_n(V)$ from V, which exists by compactness of $\mathcal{F}\mathrm{lat}_n$, gives the lower bound in (8); on the three-point grid there is one triple, $(2,0,1)$, and the displayed constant is (9). $\square$

Remark 2.7. Proposition 2.6 is the comparison [22, Proposition 5.3] at $d = n$ with exponent one and constants depending only on m and ρ. The constants come from Lemma 2.5; when the kernels entering the bounds transport the laws, the distortions are 1 and the multiplication bounds are contractions. The argument uses that $\mathcal{F}\mathrm{lat}_n$ is the graph of the map from one-step kernels to families; the model sets $\mathcal{F}_d$ of [22] for $d > n$ are closures of compressions and not graphs, and the argument does not extend to them (Open Problem (ii)). On the three-point grid the flat family generated by the one-step kernels of V lies at distance $\|\Delta_V\|$ from V, and the inequality $h_n(V) \le \|\Delta_V\|$ can be strict: Example 5.4, a family with laws that is not process-induced, has $\|\Delta_V\| = 2$ and $h_2(V) = \sqrt{2}$. Whether $h_n(V) \le \|\Delta_V\|$ holds on every grid is Open Problem (vi); Corollary 2.8 gives the constant $(m(m-1)/2)^{1/2}$ for families that transport their laws. By (8), at $d = n$ the charge and the deficit norm are comparable, with constants depending only on m and ρ, so the two carry the same information up to those constants; the content of the profile of [22] beyond the deficit norm lies at $d > n$. The coarse family of Example 3.2 has $h_2 > 0$ and is the compression of a three-state flat transport, so in the time-inhomogeneous class of [22] its profile vanishes at $d = 3$ while its deficit does not.

Corollary 2.8 (Families that transport their laws)**.** *Let $V \in \mathcal{V}_n$ carry strictly positive laws p with $V(j \leftarrow i)\, p_i = p_j$ for all $i < j$, as every process-induced family does, and let $\rho = \min_{l,v} p_l(v)$. Then*

$$\frac{\|\Delta_V\|}{\big((m-1)(2+1/\rho)\big)^{1/2}} \;\le\; h_n(V) \;\le\; \Big(\frac{m(m-1)}{2}\Big)^{1/2} \|\Delta_V\|,$$

and on the three-point grid $\{0,1,2\}$

$$\frac{\|\Delta_V\|}{\big(2+1/\min_v p_2(v)\big)^{1/2}} \;\le\; h_n(V) \;\le\; \|\Delta_V\|.$$

Proof. In the lower bound of Proposition 2.6 the kernel $K = V(r \leftarrow s)$ transports p_s to p_r, so $\varkappa_{r\leftarrow s}(V(r \leftarrow s)) = 1$ and the middle term carries the factor 1 in place of $1/\min_w p_r(w)$; the constant of the triple (t,s,r) is $2 + 1/\min_v p_t(v)$, and the sum over triples gives $(m-1)(2+1/\rho)$. The nearest flat family W need not transport the laws, and the factor $1/\min_v p_t(v)$ of the third term remains. In the upper bound each $M_k = V(k \leftarrow k{-}1)$ transports p_{k-1} to p_k, so by Lemma 2.5(a) each left multiplication is a contraction and $\|V(j \leftarrow i) - W(j \leftarrow i)\| \le \sum_{k=i+2}^{j} \|\Delta_V(k,i,k{-}1)\|$; the Cauchy–Schwarz inequality gives $\|V(j \leftarrow i) - W(j \leftarrow i)\|^2 \le (j-i-1) \sum_{k=i+2}^{j} \|\Delta_V(k,i,k{-}1)\|^2$, and summing over the pairs counts the triple $(k,i,k{-}1)$ with total weight $\sum_{j=k}^{m}(j-i-1) \le \sum_{j=2}^{m}(j-1) = m(m-1)/2$. On the three-point grid there is one triple, the sum has one term, and the constants are those displayed. □

3 Failure of monotonicity

Theorem 3.1 (The charge is not monotone under coarse-graining)**.** *Neither inequality between $h_N(V)$ and $h_n(V^\lambda)$ holds in general: there is a process-induced family with $h_N(V) = 0$ and $h_n(V^\lambda) > 0$, and a process-induced family with $h_N(V) > 0$ and $h_n(V^\lambda) = 0$. Both occur for $N \le 4$, $n = 2$, on the grid $\{0,1,2\}$, with rational data.*

The two constructions are Examples 3.2 and 3.3; all displayed numbers are exact.

Example 3.2 (Production: a lumped Markov chain). Take the three-state Markov chain with column-stochastic step and uniform initial law

$$M = \begin{pmatrix} 3/4 & 1/8 & 1/4 \\ 1/8 & 3/8 & 1/4 \\ 1/8 & 1/2 & 1/2 \end{pmatrix}, \qquad p_0 = \left(\tfrac{1}{3}, \tfrac{1}{3}, \tfrac{1}{3}\right)^{\top},$$

on the grid $\{0, 1, 2\}$, so the fine family $V(j \leftarrow i) = M^{\,j-i}$ is flat and $h_3(V) = 0$. Cut by $\lambda(1) = \lambda(2) = a$, $\lambda(3) = b$. The chain is not strongly lumpable for this partition, since the exit probabilities into $\{3\}$ from states 1 and 2 are $1/8$ and $1/2$; the positive charge of the coarse family follows from the nonzero deficit computed below and not from this. The coarse laws are $p_0^\lambda = (\frac{2}{3}, \frac{1}{3})$, $p_1^\lambda = (\frac{5}{8}, \frac{3}{8})$, $p_2^\lambda = (\frac{41}{64}, \frac{23}{64})$, and the coarse family is

$$V^\lambda(1 \leftarrow 0) = \begin{pmatrix} 11/16 & 1/2 \\ 5/16 & 1/2 \end{pmatrix}, \quad V^\lambda(2 \leftarrow 1) = \begin{pmatrix} 29/40 & 1/2 \\ 11/40 & 1/2 \end{pmatrix}, \quad V^\lambda(2 \leftarrow 0) = \begin{pmatrix} 85/128 & 19/32 \\ 43/128 & 13/32 \end{pmatrix}.$$

The single deficit is

$$\Delta_{V^\lambda}(2, 0, 1) = \begin{pmatrix} 3/320 & -3/160 \\ -3/320 & 3/160 \end{pmatrix}, \qquad \|\Delta_{V^\lambda}\|^2 = \frac{18}{23575}, \tag{10}$$

in the metric (6) with the coarse laws. Since $\mathcal{F}\mathrm{lat}_2$ is the zero set of the deficit, $h_2(V^\lambda) > 0$. The coarse family is process-induced, so by the three-point bound of Corollary 2.8 with $\min p_2^\lambda = 23/64$ the constant is $(2 + 64/23)^{1/2} = (110/23)^{1/2}$, and

$$\sqrt{\tfrac{9}{56375}} \;\le\; h_2(V^\lambda) \;\le\; \sqrt{\tfrac{18}{23575}},$$

that is, after outward rounding, $0.0126 \le h_2(V^\lambda) \le 0.0277$. The fine family has charge 0 and the coarse family has positive charge.

Example 3.3 (Destruction: discarding the memory-bearing factor). Let B denote the charged two-state family of Example 3.2, $B = V^\lambda$ with laws $q_i = p_i^\lambda$; it is process-induced, being the observed family of the lumped chain. Let A be the flat two-state family of the symmetric chain $M_A = \left(\begin{smallmatrix} 3/4 & 1/4 \\ 1/4 & 3/4 \end{smallmatrix}\right)$ at its stationary law $(\frac{1}{2}, \frac{1}{2})$. Run the two processes independently: the pair process on $[2] \times [2] \cong [4]$ has the product $V = A \otimes B$ as its conditional family, with product laws, and its deficit factorizes through flatness of A:

$$\Delta_V(2, 0, 1) \;=\; A(2 \leftarrow 0) \otimes \Delta_B(2, 0, 1), \qquad \|\Delta_V\|^2 = \|A(2 \leftarrow 0)\|^2 \cdot \|\Delta_B\|^2 = \frac{17}{16} \cdot \frac{18}{23575} = \frac{153}{188600},$$

where $\|A(2 \leftarrow 0)\|^2$ is the law-metric size of the kernel $A(2 \leftarrow 0)$ itself, $\sum_{v,u} p_0'(u) A(2 \leftarrow 0)_{vu}^2 / p_2'(v)$. So $\|\Delta_V\| > 0$ and $h_4(V) > 0$; the product family is process-induced, so by the three-point bound of Corollary 2.8 with the product law at time 2, whose minimum is $23/128$, the constant is $(2 + 128/23)^{1/2} = (174/23)^{1/2}$ and $\sqrt{51/475600} \le h_4(V) \le \sqrt{153/188600}$. Cut by the first-coordinate projection $\lambda(v_A, v_B) = v_A$. The compression returns the A family, $V^\lambda = A$, which is flat: $h_2(V^\lambda) = 0$. The discarded factor is autonomous and carries the whole deficit. Proposition 6.4 shows that the flux of the fine deficit through this cut vanishes identically, so the deficit is severed in full.

Example 3.4 (A closed cut). Replace M in Example 3.2 by

$$M_{\mathrm{sl}} = \begin{pmatrix} 1/2 & 1/4 & 1/8 \\ 1/4 & 1/2 & 1/8 \\ 1/4 & 1/4 & 3/4 \end{pmatrix},$$

which is strongly lumpable for the same partition: the exit mass into $\{3\}$ equals $1/4$ from both states of the block. The coarse family is flat. The same cut produces charge in Example 3.2 and none here; Theorem 4.5 covers both instances by one criterion.

4 The balance law

4.1 Flux, production, and the severed component

Definition 4.1. Let (V, p) be a family on $[N]$ with strictly positive laws and λ a cut. For $s < r < t$ define

$$\begin{aligned} \text{the } \textit{flux} \qquad & \Phi_\lambda(t,s,r) \;=\; \mathcal{C}_s\big[\Delta_V(t,s,r)\big] \;=\; L\,\Delta_V(t,s,r)\,R_s, \\ \text{the } \textit{production} \qquad & \mathrm{P}_\lambda(t,s,r) \;=\; L\,V(t \leftarrow r)\,Q_r\,V(r \leftarrow s)\,R_s, \\ \text{the } \textit{severed deficit} \qquad & \Xi_\lambda(t,s,r) \;=\; \Delta_V(t,s,r) \;-\; \iota_s\big[\Phi_\lambda(t,s,r)\big]. \end{aligned}$$

All three have vanishing column sums. Flux and production are coarse objects measured in the induced metric; the severed deficit is a fine object. Norms of cochains are the triple sums of Section 2. The production term is a kernel difference; its magnitude is $\|\mathrm{P}_\lambda\|$, and it has no sign.

Theorem 4.2 (Balance law)**.** *For every cut λ and all $s < r < t$:*

(i) (Balance) $\Delta_{V^\lambda}(t,s,r) \;=\; \Phi_\lambda(t,s,r) \;+\; \mathrm{P}_\lambda(t,s,r)$*, an identity of matrices.*

(ii) (Conservation across the cut) $\|\Delta_V\|^2 \;=\; \|\Phi_\lambda\|^2 \;+\; \|\Xi_\lambda\|^2$ *in the metric* (6)*: the fine deficit splits orthogonally into its transmitted and severed components. In particular $\|\Phi_\lambda\| \le \|\Delta_V\|$, with equality if and only if $\Xi_\lambda = 0$.*

(iii) (A priori bounds) $\big|\,\|\mathrm{P}_\lambda\| - \|\Delta_{V^\lambda}\|\,\big| \;\le\; \|\Phi_\lambda\| \;\le\; \|\Delta_V\|$.

Proof. (i) Insert $I = \Pi_r + Q_r$ at the cut time:

$$\begin{aligned} \Delta_{V^\lambda}(t,s,r) &= LV(t \leftarrow s)R_s - LV(t \leftarrow r)R_r\,LV(r \leftarrow s)R_s \\ &= L\big[V(t \leftarrow s) - V(t \leftarrow r)\Pi_r V(r \leftarrow s)\big]R_s, \end{aligned}$$

and $V(t \leftarrow s) - V(t \leftarrow r)\Pi_r V(r \leftarrow s) = \Delta_V(t,s,r) + V(t \leftarrow r)Q_r V(r \leftarrow s)$. (ii) is Lemma 2.3(c) applied triple by triple, with the isometry of ι_s identifying $\|\mathcal{C}_s[\Delta_V]\|$ in the induced metric with the norm of the block-measurable component. (iii) is the triangle inequality applied to (i), then (ii). □

Remark 4.3. Part of the fine deficit, Ξ_λ, is orthogonal to the block-measurable subspace for the cut λ in the metric (6); it is invisible to observables measurable with respect to that partition, not to every coarse observation, and the compression discards it. By Lemma 2.3, $\|\Xi_\lambda(t,s,r)\|^2 = \mathbb{E}_{p_t \otimes p_s}\big[\mathrm{Var}(f_{\Delta_V(t,s,r)} \mid \mathcal{B}_\lambda)\big]$: the squared norm of the severed component is the mean conditional variance of the deficit density given the coarse partition. The rest, Φ_λ, is transmitted without amplification. The coarse deficit beyond the transmitted part is the production term, which involves the family only through its passage across the fluctuation subspace at the cut time. In Example 3.2 the fine deficit vanishes, so flux and severed part vanish and $\Delta_{V^\lambda} = \mathrm{P}_\lambda$, which (10) displays. In Example 3.3 the production vanishes by Proposition 6.4 and the flux vanishes, so the whole deficit is severed. Section 7 lists the four families of the paper by mechanism.

4.2 The zero set of production

Definition 4.4. The cut λ is *dynamically closed* for (V, p) at level r if

$$L\,V(t \leftarrow r)\,Q_r \;=\; 0 \qquad \text{for every } t > r,$$

that is, if the conditional law of the coarse future given the fine present, $u \mapsto \big(LV(t \leftarrow r)\big)_{\cdot u}$, is constant on the blocks of λ; it is dynamically closed if it is closed at every level $0 < r < m$. The *accessible subspace* at level r is the linear span $\mathcal{A}_r = \operatorname{span}\{\, V(r \leftarrow s)R_s\, e_a :\ s < r,\ a \in [n] \,\} \subseteq \mathbb{R}^N$. The cut is *closed on the accessible subspace* at level r if $LV(t \leftarrow r)Q_r$ vanishes on $\mathcal{A}_r$ for every $t > r$. Dynamical closure implies closure on the accessible subspace, and the converse can fail when $\mathcal{A}_r \neq \mathbb{R}^N$ (Example 4.7).

Theorem 4.5 (Zero set of production)**.** *For every cut λ:*

(i) if λ is dynamically closed, then $\mathrm{P}_\lambda \equiv 0$;

(ii) $\mathrm{P}_\lambda \equiv 0$ if and only if λ is closed on the accessible subspace at every level $0 < r < m$.

Proof. (i) Each production term contains the factor $LV(t \leftarrow r)Q_r$ on the left of $V(r \leftarrow s)R_s$. (ii) The a-th column of $\mathrm{P}_\lambda(t, s, r)$ is $LV(t \leftarrow r)Q_r$ applied to the vector $V(r \leftarrow s)R_s\, e_a$, and as $s < r$ and $a \in [n]$ vary these vectors span $\mathcal{A}_r$ by definition; so $\mathrm{P}_\lambda(t, s, r) = 0$ for all $s < r < t$ with cut time r if and only if $LV(t \leftarrow r)Q_r$ vanishes on $\mathcal{A}_r$ for all $t > r$. □

Corollary 4.6 (Two-time Chapman–Kolmogorov lumpability)**.** *Let X be a process with strictly positive one-time laws whose conditional family V is flat, as is the case for a Markov chain, and let λ be a cut. Call (X, λ)* two-time Chapman–Kolmogorov lumpable *on the grid if the two-time conditionals of $Y = \lambda(X)$ satisfy* (1)*. The following are equivalent:*

(i) $\mathrm{P}_\lambda \equiv 0$;

(ii) V^λ is flat, that is, $h_n(V^\lambda) = 0$;

(iii) (X, λ) is two-time Chapman–Kolmogorov lumpable.

Each is implied by weak lumpability, Markovianity of Y for the given initial law [8, 17], and by strong lumpability of the one-step kernels for the partition, $LM_k = \mathsf{m}_k L$ for every k and some column-stochastic m_k, where $M_k = V(k \leftarrow k{-}1)$. Neither implication reverses: Example 4.7 gives a lumped process that is two-time Chapman–Kolmogorov lumpable and not Markov, so (iii) implies neither weak nor strong lumpability, and a chain can be weakly lumpable for one initial law without being strongly lumpable [8]. Processes with composing two-time kernels that are not Markov are classical [6].

Proof. Flatness of V kills the flux, so $\Delta_{V^\lambda} = \mathrm{P}_\lambda$ by Theorem 4.2(i), which gives (i)⇔(ii); (ii)⇔(iii) is Proposition 2.2 and the definition of flatness. If Y is Markov, its conditional family composes, which is (iii). For strong lumpability [8], $LM_k = \mathsf{m}_k L$ for all k gives $LV(t \leftarrow r) = LM_t \cdots M_{r+1} = \mathsf{m}_t \cdots \mathsf{m}_{r+1} L$, and $LQ_r = L - LR_rL = L - L = 0$, so $LV(t \leftarrow r)Q_r = \mathsf{m}_t \cdots \mathsf{m}_{r+1} LQ_r = 0$: the cut is dynamically closed. Example 3.4 is this case. The non-implications are Example 4.7 and the weakly but not strongly lumpable chains of [8]; the structure of the weak-lumpability set is treated in [17, 16]. □

Example 4.7 (Two-time composition without Markovianity)*.* Let Y_0 and Y_1 be independent fair bits, let $Y_2 = Y_0 \oplus Y_1$ be their sum modulo 2, and let $X_0 = (Y_0, Y_1)$, $X_1 = (Y_1, Y_0)$, $X_2 = (Y_0 \oplus Y_1,\, Y_1)$ on $[2] \times [2] \cong [4]$. Each step is a bijection of the state space, so X is a Markov chain with permutation one-step kernels, uniform laws, and a flat family; the cut λ is the first coordinate, and $Y = \lambda(X)$. The bits Y_0, Y_1, Y_2 are pairwise independent, so every coarse kernel is $J/2$, the kernel with all entries $\frac{1}{2}$; the coarse family composes, $\Delta_{V^\lambda} = 0$, and $\mathrm{P}_\lambda = 0$. The lumped process is not Markov: $\mathbb{P}(Y_2 = 0 \mid Y_1 = 0, Y_0 = 0) = 1$ and $\mathbb{P}(Y_2 = 0 \mid Y_1 = 0) = \frac{1}{2}$. The cut is closed on the accessible

subspace at level 1 and is not dynamically closed there: in the state order $(0,0),(0,1),(1,0),(1,1)$, the rows of $LV(2 \leftarrow 1)Q_1$ are $\pm(\frac{1}{2},-\frac{1}{2},-\frac{1}{2},\frac{1}{2})$, and $\mathcal{A}_1$ is spanned by $\frac{1}{2}(1,0,1,0)^\top$ and $\frac{1}{2}(0,1,0,1)^\top$, on which those rows vanish.

Corollary 4.8 (Charge-level bounds from the balance law)**.** *Let L_* and C_* be the constants of Proposition 2.6 for the coarse laws p^λ at dimension n, so that $C_* = 1$ on the three-point grid. Then*

$$\frac{\big(\|\mathrm{P}_\lambda\| - \|\Phi_\lambda\|\big)_+}{L_*} \;\le\; \frac{\|\Phi_\lambda + \mathrm{P}_\lambda\|}{L_*} \;\le\; h_n(V^\lambda) \;\le\; C_*\,\|\Phi_\lambda + \mathrm{P}_\lambda\| \;\le\; C_*\big(\|\Phi_\lambda\| + \|\mathrm{P}_\lambda\|\big),$$

with $\|\Phi_\lambda\| \le \|\Delta_V\|$, and for a flat fine family

$$\frac{\|\mathrm{P}_\lambda\|}{L_*} \;\le\; h_n(V^\lambda) \;\le\; C_*\,\|\mathrm{P}_\lambda\|.$$

These are bounds; the charge does not satisfy an additive identity of its own.

Proof. Theorem 4.2(i) gives $\Delta_{V^\lambda} = \Phi_\lambda + \mathrm{P}_\lambda$; Proposition 2.6 applied to V^λ and the triangle inequality give the rest. □

Proposition 4.9 (Dependence on the long kernel)**.** *Fix $s < r < t$, the cut, the laws, and the kernels $V(t \leftarrow r)$ and $V(r \leftarrow s)$. Then $\mathrm{P}_\lambda(t,s,r)$ does not depend on $V(t \leftarrow s)$, and the map $V(t \leftarrow s) \mapsto \Phi_\lambda(t,s,r) = \mathcal{C}_s[V(t \leftarrow s)] - \mathcal{C}_s[V(t \leftarrow r)V(r \leftarrow s)]$ is affine. Hence, for fixed short kernels, the condition $\langle \Phi_\lambda(t,s,r), \mathrm{P}_\lambda(t,s,r)\rangle \ge 0$ is a closed half-space condition on $\mathcal{C}_s[V(t \leftarrow s)]$, and $\|\Delta_{V^\lambda}(t,s,r)\|^2 = \|\Phi_\lambda(t,s,r) + \mathrm{P}_\lambda(t,s,r)\|^2$ is a convex quadratic function of $V(t \leftarrow s)$.*

Proof. This follows from the definitions of P_λ and Φ_λ in Definition 4.1. □

5 Production as a Mori–Zwanzig memory kernel

The production term has the form of a memory kernel: resolved propagation to the cut, a fluctuation insertion, resolved propagation after.

5.1 The discrete Dyson expansion

Proposition 5.1 (Resolved and fluctuation segments of the coarse family)**.** *Let V be flat with one-step kernels $M_k = V(k \leftarrow k-1)$ and let λ be a cut. For $a < b$ define the* segment kernels

$$G(b \leftarrow a) \;=\; L\,M_b\,Q_{b-1}\,M_{b-1}\,Q_{b-2}\cdots Q_{a+1}\,M_{a+1}\,R_a,$$

so that $G(a{+}1 \leftarrow a) = \mathsf{m}_{a+1} = LM_{a+1}R_a$ is the coarse one-step kernel. Then for all $i < j$,

$$V^\lambda(j \leftarrow i) \;=\; \sum_{i=a_0<a_1<\cdots<a_k=j} G(a_k \leftarrow a_{k-1}) \cdots G(a_1 \leftarrow a_0). \tag{11}$$

The word with all segments of length one is the flat part $\mathsf{m}_j \cdots \mathsf{m}_{i+1}$; every longer segment carries at least one fluctuation insertion. On the three-point window $s < r < t$ the deficit is the single-insertion kernel:

$$\Delta_{V^\lambda}(t,s,r) \;=\; L\,V(t \leftarrow r)\,Q_r\,V(r \leftarrow s)\,R_s \;=\; \mathrm{P}_\lambda(t,s,r). \tag{12}$$

Proof. Write $V^\lambda(j \leftarrow i) = LM_j\Big(\prod_{k=j-1}^{i+1}(\Pi_k+Q_k)M_k\Big)R_i$ and expand multilinearly over the choices Π_k versus Q_k. Each occurrence of $\Pi_k = R_kL$ splits its word into two factors joined at time k; grouping the factors between consecutive Π's and the endpoints yields the segment kernels, and the sum over binary words is the sum over compositions $i = a_0 < \cdots < a_k = j$ of the interval. Equation (12) is Theorem 4.2(i) with vanishing flux. □

Equation (11) is an algebraic identity on the grid. In the correspondence with the projection-operator formalism [12, 27, 11, 7, 28], in the discrete-time form used by optimal prediction and by data-driven model reduction [3, 9], the all-Π word is the resolved evolution of the coarse family, each run of Q's is a segment of orthogonal dynamics, and the segment kernels of length two or more are the memory kernels. The objects here are two-time conditional kernels, not observables, so the formalism's fluctuating force has no counterpart, and no continuous-time limit is taken; the correspondence is with the discrete-time projected dynamics and not an equivalence with the continuous-time generalized Langevin equation. The elimination-of-variables reading is classical [24]. The deficit (12) is the single-insertion term at the level of two-time kernels: the *fluctuation current* across the cut, the part of the $s \to t$ transition current that crosses the cut time inside the subspace invisible to the coarse description. Production sources the failure of two-time divisibility; its comparison with the information backflow of open quantum systems is an analogy, as stated in the positioning paragraph of Section 1, and no relation between P_λ and a distinguishability functional is claimed.

5.2 The law section among admissible sections

The Mori–Zwanzig formalism begins with a choice of projection: Zwanzig's conditional expectation, Mori's linear projection onto a span of observables, the Wiener projections of [9]. Different choices give different memory kernels for the same reduction. The following theorem states what is independent of the choice in the present setting, and its scope: the deficit of the observed family is defined without a projection, and the class over which uniqueness is proved is the class of block-supported stochastic sections, which contains neither Mori's linear projections nor the Wiener projections.

Theorem 5.2 (The law section)**.** *Let V be flat with laws p and λ a cut.*

(i) The deficit Δ_{V^λ} and the charge $h_n(V^\lambda)$ are functionals of the observed family V^λ alone; no projection enters their definition.

(ii) The memory kernel built from the law projection Π_r equals the deficit: (12)*.*

(iii) Let $\widetilde{R}_r$ be any admissible section ($L\widetilde{R}_r = I_n$, columns non-negative and supported on the blocks), with idempotent $\widetilde{\Pi}_r = \widetilde{R}_rL$ and kernel $\widetilde{\mathrm{P}}(t,s,r) = LV(t \leftarrow r)(I - \widetilde{\Pi}_r)V(r \leftarrow s)R_s$. Then

$$\widetilde{\mathrm{P}}(t,s,r) \;-\; \Delta_{V^\lambda}(t,s,r) \;=\; L\,V(t \leftarrow r)\,\big(R_r - \widetilde{R}_r\big)\,V^\lambda(r \leftarrow s),$$

and, with the time-r law and hence R_r held fixed, the difference vanishes for every flat family with that time-r law if and only if $\widetilde{R}_r = R_r$. The law section is thus the unique admissible section whose kernel equals the deficit for every flat family with the given time-r law.

Proof. (i) is the definition of the deficit and of (3). (ii) is (12). (iii): subtract $\Delta_{V^\lambda} = LV(t \leftarrow r)(I - R_rL)V(r \leftarrow s)R_s$ from $\widetilde{\mathrm{P}}$ and use $LV(r \leftarrow s)R_s = V^\lambda(r \leftarrow s)$; the difference is $LV(t \leftarrow r)(R_r - \widetilde{R}_r)\,V^\lambda(r \leftarrow s)$. For the uniqueness claim, $L(R_r - \widetilde{R}_r) = 0$ always, so the discrepancy is invisible to families that do not separate states within a block. If $\widetilde{R}_r \neq R_r$, pick a block a on which the sections differ and a state u_0 of that block with $d(u_0) \neq 0$, where $d = (R_r - \widetilde{R}_r)e_a$, a

nonzero vector supported on the block with $\sum_u d(u) = 0$. Build a flat family, carrying the given law at every time up to r and strictly positive laws afterwards, from one-step kernels: the identity on every step up to time r, so that $V(r \leftarrow s) = I$ and $V^\lambda(r \leftarrow s) = I_n$, and, on the step leaving r, the kernel sending u_0 into one coarse block b and every other state of $[N]$ into a different block b'; then, at $t = r + 1$, the b-entry of the a-th column of the difference is $\big(LV(t \leftarrow r)\, d\big)_b = d(u_0) \neq 0$. The detecting kernels may be taken strictly positive by convex perturbation toward a reset kernel $w\mathbf{1}^\top$ with $w > 0$; the reset kernel annihilates d since $\mathbf{1}^\top d = 0$, so the perturbation preserves the nonvanishing of the difference. □

Remark 5.3 (The residual at a nearest flat family). Let $W^* \in \mathcal{F}\mathrm{lat}_n$ attain $h_n(V^\lambda)$; such a W^* exists by compactness. The residual $V^\lambda - W^*$ is normal to the flat locus at W^* in the sense of the first-order theory of [22, Section 6] and the normal cones of variational analysis [15], and it is defined without a projection. Its norm is $h_n(V^\lambda)$, and for a flat fine family, by (12) and Proposition 2.6,

$$\frac{\|\mathrm{P}_\lambda\|}{L_*} \;\le\; \|V^\lambda - W^*\| \;=\; h_n(V^\lambda) \;\le\; C_*\,\|\mathrm{P}_\lambda\|,$$

with L_* and C_* the constants of Proposition 2.6 for the coarse laws. The norm $h_n(V^\lambda)$ is canonical and the residual is not: $\mathcal{F}\mathrm{lat}_n$ is the zero set of a polynomial system and is not convex, a nearest flat family need not be unique in general, and distinct nearest families have distinct residuals of the same norm (Example 5.4, a family with laws that is not process-induced). The set of families with more than one nearest flat family has Lebesgue measure zero on the affine hull of $\mathcal{V}_n$: $\mathcal{F}\mathrm{lat}_n$ is closed, and for every closed subset of a Euclidean space the points with more than one nearest point lie in countably many surfaces of finite measure of codimension one and form a Lebesgue-null set [5, Theorem 1]. Whether a process-induced family can have more than one nearest flat family is Open Problem (vii). The statements of this paper use only h_n.

Example 5.4 (Two nearest flat families). Take $n = 2$, the grid $\{0, 1, 2\}$, uniform laws at all three times, so that (6) is the sum of squares of entries, and the family

$$V(1 \leftarrow 0) = V(2 \leftarrow 1) = I, \qquad V(2 \leftarrow 0) = \sigma = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

This family carries uniform laws and is not process-induced, since no joint law has the identity coupling at the pairs $(0, 1)$ and $(1, 2)$ and the swap coupling at $(0, 2)$. Its deficit is $\sigma - I$, of squared norm 4. A flat family is generated by one-step kernels $A = \left(\begin{smallmatrix} a_1 & a_2 \\ 1-a_1 & 1-a_2 \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} b_1 & b_2 \\ 1-b_1 & 1-b_2 \end{smallmatrix}\right)$, and

$$\mathrm{dist}(V, W)^2 \;=\; \|I - A\|^2 + \|I - B\|^2 + \|\sigma - BA\|^2 \;=\; 2\big(x^2 + y^2 + u^2 + v^2 + c_{11}^2 + c_{22}^2\big),$$

with $x = 1 - a_1$, $y = a_2$, $u = 1 - b_1$, $v = b_2$ in $[0, 1]$ and $c_{11} = (BA)_{11} = (1 - x)(1 - u) + xv$, $c_{22} = (BA)_{22} = (1-y)(1-v)+yu$. Put $\alpha = x+y$ and $\beta = u+v$. Then $c_{11}+c_{22} = 1+(1-\alpha)(1-\beta)$, and by the inequalities $x^2 + y^2 \ge \alpha^2/2$, $u^2 + v^2 \ge \beta^2/2$, and $c_{11}^2 + c_{22}^2 \ge (c_{11} + c_{22})^2/2$,

$$x^2 + y^2 + u^2 + v^2 + c_{11}^2 + c_{22}^2 \;\ge\; \frac{\alpha^2 + \beta^2 + \big(1 + (1-\alpha)(1-\beta)\big)^2}{2} \;=\; 1 + \frac{(1-\varsigma)^2 + \tau^2}{2},$$

where $\varsigma = (1 - \alpha) + (1 - \beta)$ and $\tau = (1 - \alpha)(1 - \beta)$. So $\mathrm{dist}(V, W)^2 \ge 2$, with equality if and only if $\tau = 0$, $\varsigma = 1$, $x = y$, $u = v$, and $c_{11} = c_{22}$, that is, $(x, y, u, v) = (\frac{1}{2}, \frac{1}{2}, 0, 0)$ or $(0, 0, \frac{1}{2}, \frac{1}{2})$. Hence $h_2(V) = \sqrt{2}$, attained at exactly two flat families, W' with one-step kernels $(J/2,\, I)$ and W'' with one-step kernels $(I,\, J/2)$, where $J/2$ is the kernel with all entries $1/2$. The residuals are $V - W' = (I - J/2,\ 0,\ \sigma - J/2)$ and $V - W'' = (0,\ I - J/2,\ \sigma - J/2)$ on the pairs $(0, 1)$, $(1, 2)$, $(0, 2)$; they differ by $1/2$ in every entry of the first two pairs. The upper bound of (9) is $\|\Delta_V\| = 2$, and it is strict here.

6 Towers and monotonicity

6.1 Towers and the telescoped balance law

A *tower* is a sequence of cuts

$$[N] = [n_0] \xrightarrow{\lambda_1} [n_1] \xrightarrow{\lambda_2} \cdots \xrightarrow{\lambda_K} [n_K], \qquad n_0 > n_1 > \cdots > n_K \geq 2,$$

with families $V^{(0)} = V$ and $V^{(k)} = \big(V^{(k-1)}\big)^{\lambda_k}$, laws lumped accordingly. Disintegrations compose, $R_i^{\mu\circ\lambda} = R_i^{\lambda} R_i^{\mu}$ by direct computation on the nested blocks, so $L_\mu(L_\lambda V R_i^\lambda) R_i^\mu = L_{\mu\circ\lambda} V R_i^{\mu\circ\lambda}$: the tower and the composed cut give the same family.

Corollary 6.1 (Telescoped balance)**.** *Writing $\mathcal{C}^{(k)}$, $\mathrm{P}^{(k)}$, $\Phi^{(k)}$, $\Xi^{(k)}$ for the objects of Definitions 2.1 and 4.1 at step k,*

$$\Delta_{V^{(K)}} \;=\; \mathcal{C}^{(K)} \cdots \mathcal{C}^{(1)} \big[\Delta_V\big] \;+\; \sum_{k=1}^{K} \mathcal{C}^{(K)} \cdots \mathcal{C}^{(k+1)} \big[\mathrm{P}^{(k)}\big],$$

each compression contractive in its induced metric.

Proof. The statement follows by induction on K from Theorem 4.2(i). □

6.2 Monotonicity of the deficit norm

Theorem 6.2 (Monotonicity of the deficit norm)**.** *Consider one tower step λ applied to (V, p), and suppose*

(S) *nothing is severed: $\Xi_\lambda = 0$, equivalently $\|\Phi_\lambda\| = \|\Delta_V\|$; and*

(C) *production is coherent: $\langle \Phi_\lambda, \mathrm{P}_\lambda \rangle \geq 0$ as cochains in the induced metric.*

Then

$$\|\Delta_{V^\lambda}\|^2 \;=\; \|\Delta_V\|^2 \;+\; 2\langle \Phi_\lambda, \mathrm{P}_\lambda \rangle \;+\; \|\mathrm{P}_\lambda\|^2 \;\geq\; \|\Delta_V\|^2 + \|\mathrm{P}_\lambda\|^2,$$

so the deficit norm does not decrease, and it increases strictly when $\mathrm{P}_\lambda \neq 0$. Along a tower every step of which satisfies (S) *and* (C)*, the deficit norm is nondecreasing in k. The charge satisfies, by Proposition 2.6 with the constants $L_{*,k}$, $C_{*,k}$ of the laws at level k,*

$$h_{n_{k+1}}\big(V^{(k+1)}\big) \;\geq\; \frac{\|\Delta_{V^{(k+1)}}\|}{L_{*,k+1}} \;\geq\; \frac{\|\Delta_{V^{(k)}}\|}{L_{*,k+1}} \;\geq\; \frac{h_{n_k}\big(V^{(k)}\big)}{C_{*,k} L_{*,k+1}};$$

this chain does not imply monotonicity of the charge, since $C_{*,k} L_{*,k+1} > 1$, and positivity of the charge is preserved. Both hypotheses are sharp: Example 3.3 violates (S)*, satisfies* (C) *trivially since its production vanishes, and has a flat coarse family; Proposition 6.5 violates* (C)*, satisfies* (S) *exactly, and has a flat coarse family.*

Proof. By Theorem 4.2(i), $\|\Delta_{V^\lambda}\|^2 = \|\Phi_\lambda + \mathrm{P}_\lambda\|^2 = \|\Phi_\lambda\|^2 + 2\langle \Phi_\lambda, \mathrm{P}_\lambda \rangle + \|\mathrm{P}_\lambda\|^2$; under (S), $\|\Phi_\lambda\|^2 = \|\Delta_V\|^2$ by Theorem 4.2(ii), and (C) fixes the sign of the cross term. The tower statement iterates, and the charge chain is Proposition 2.6 applied at levels $k+1$ and k. □

Remark 6.3. The identity in the theorem holds without hypotheses and locates the ways the deficit norm can fall in one step: the cut severs deficit-bearing directions (failure of (S)), or production interferes destructively with the transmitted deficit (failure of (C)). The first mechanism is the general form of discarding an autonomous factor (Proposition 6.4); the second is interference at the cut, and Proposition 6.5 shows it can act alone and cancel the deficit entirely. Any increase of the coarse deficit norm beyond the transmitted fine deficit is supplied by the production term: with $\mathrm{P}_\lambda = 0$, $\Delta_{V^\lambda} = \Phi_\lambda$ and $\|\Phi_\lambda\| \le \|\Delta_V\|$. Nonzero production does not by itself increase the norm; whether the norm increases depends on the sign of the cross term, and Proposition 4.9 describes the dependence of that sign on the long kernel.

6.3 Autonomous factors and total severing

Proposition 6.4 (Discarding an autonomous factor)**.** *Let the fine process be an independent pair, $X = (X', X'')$ on $[n'] \times [n'']$, so that $V = V' \otimes V''$ with product laws $p_i = p'_i \otimes p''_i$, and let $\lambda(v', v'') = v'$ be the projection discarding the autonomous factor X''. Then:*

(i) the cut is dynamically closed: $\mathrm{P}_\lambda \equiv 0$;

(ii) the flux is the retained factor's own deficit: $\Phi_\lambda(t, s, r) = \Delta_{V'}(t, s, r)$, and $V^\lambda = V'$;

(iii) hence $\|\Xi_\lambda\|^2 = \|\Delta_V\|^2 - \|\Delta_{V'}\|^2$: the deficit of the discarded factor, including its cross terms with the retained factor, is severed in full. If the retained factor is flat and the discarded factor carries deficit, the flux vanishes identically and the whole deficit is severed, which is Example 3.3.

Proof. With product laws, $\Pi_r = I_{n'} \otimes p''_r \mathbf{1}^\top$ and $Q_r = I_{n'} \otimes (I_{n''} - p''_r \mathbf{1}^\top)$. (i) $LV(t \leftarrow r)Q_r = V'(t \leftarrow r) \otimes \big[\mathbf{1}^\top V''(t \leftarrow r)(I - p''_r \mathbf{1}^\top)\big]$, and $\mathbf{1}^\top V''(t \leftarrow r) = \mathbf{1}^\top$ gives $\mathbf{1}^\top (I - p''_r \mathbf{1}^\top) = \mathbf{1}^\top - \mathbf{1}^\top = 0$. (ii) For any kernels A on $[n']$ and B on $[n'']$, $\mathcal{C}_s[A \otimes B] = A \cdot \big(\sum_u p''_s(u) \sum_v B_{vu}\big)$; the deficit of a product with laws splits as $\Delta_V = \Delta_{V'} \otimes V''(t \leftarrow s) + \big(V'(t \leftarrow r)V'(r \leftarrow s)\big) \otimes \Delta_{V''}$; the first factor's companion $V''(t \leftarrow s)$ has unit column sums and the second's companion $\Delta_{V''}$ has zero column sums, so the compression returns $\Delta_{V'} \cdot 1 + \big(V'V'\big) \cdot 0$. (iii) is Theorem 4.2(ii) with (ii). □

A tower discards an autonomous factor when some step projects away an independent subdynamics; then hypothesis (S) fails by the discarded factor's share of the deficit, and Example 3.3 realizes the failure with all quantities exact.

6.4 Perfect interference: sharpness of the coherence hypothesis

Hypothesis (S) does not suffice on its own. The following construction gives a process whose cut severs nothing, is not an autonomous-factor projection, and has production term equal to the negative of the transmitted deficit, so that the two cancel and the coarse family is flat.

Proposition 6.5 (Perfect interference)**.** *There is a process-induced family on three states over the grid $\{0, 1, 2\}$, with an explicitly displayed strictly positive joint law, and a cut, such that*

(i) nothing is severed: $\Xi_\lambda = 0$; and the cut is not an autonomous-factor projection of the type of Proposition 6.4, since $\mathrm{P}_\lambda \neq 0$ while such projections have $\mathrm{P}_\lambda \equiv 0$;

(ii) production is nonzero and maximally incoherent: $\mathrm{P}_\lambda = -\Phi_\lambda$, so $\langle \Phi_\lambda, \mathrm{P}_\lambda \rangle = -\|\Phi_\lambda\| \, \|\mathrm{P}_\lambda\| < 0$;

(iii) hence $\Delta_{V^\lambda} = \Phi_\lambda + \mathrm{P}_\lambda = 0$: the coarse family is flat, and the cut cancels the two-time composition defect of the fine family entirely.

Thus hypothesis (C) *cannot be dropped from Theorem 6.2 even when* (S) *holds exactly.*

Proof. All laws are uniform, $p_0 = p_1 = p_2 = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})^\top$, and the cut is $\lambda(1) = \lambda(2) = a$, $\lambda(3) = b$, with coarse laws $(\frac{2}{3}, \frac{1}{3})$. Take the doubly stochastic one-step kernels

$$A = V(1 \leftarrow 0) = \begin{pmatrix} 1/2 & 1/4 & 1/4 \\ 1/4 & 1/4 & 1/2 \\ 1/4 & 1/2 & 1/4 \end{pmatrix}, \qquad B = V(2 \leftarrow 1) = \begin{pmatrix} 1/4 & 1/4 & 1/2 \\ 1/4 & 1/2 & 1/4 \\ 1/2 & 1/4 & 1/4 \end{pmatrix},$$

and perturb the long kernel off the Markov product by a block lift: with

$$\delta_0 = \begin{pmatrix} 1 & -2 \\ -1 & 2 \end{pmatrix}, \qquad \iota_0[\delta_0] = \begin{pmatrix} 1/2 & 1/2 & -1 \\ 1/2 & 1/2 & -1 \\ -1 & -1 & 2 \end{pmatrix},$$

$$V(2 \leftarrow 0) \;=\; BA + \tfrac{1}{64}\,\iota_0[\delta_0] = \begin{pmatrix} 41/128 & 49/128 & 19/64 \\ 41/128 & 41/128 & 23/64 \\ 23/64 & 19/64 & 11/32 \end{pmatrix},$$

which is entrywise positive and column-stochastic, and preserves the uniform law, since $\iota_0[\delta_0]$ has vanishing column and row sums. The line $\mathbb{R}\,\delta_0$ is the set of coarse kernel differences with vanishing column sums whose block lifts preserve the uniform law for this partition, so the construction stays inside marginal-consistent families.

The family is process-induced. Define the joint law on $[3]^3$ by

$$\pi(u, v, w) \;=\; \tfrac{1}{3}\,A_{vu}B_{wv} \;+\; \tfrac{1}{9} \cdot \tfrac{1}{64}\,\iota_0[\delta_0]_{wu} \;\geq\; \tfrac{1}{48} - \tfrac{1}{576} \;=\; \tfrac{11}{576} \;>\; 0.$$

Summing over w kills the correction column-wise and returns the $(0,1)$ margin $\frac{1}{3}A_{vu}$; summing over u kills it row-wise and returns the $(1,2)$ margin $\frac{1}{3}B_{wv}$; summing over v returns $\frac{1}{3}\big(BA + \frac{1}{64}\iota_0[\delta_0]\big)_{wu}$, the $(0,2)$ margin. So the three two-time conditional kernels of the process with law π are A, B, and $V(2 \leftarrow 0)$, with uniform one-time laws.

Nothing is severed. The single fine deficit is $\Delta_V(2,0,1) = V(2 \leftarrow 0) - BA = \frac{1}{64}\iota_0[\delta_0]$, a block lift; hence $\Phi_\lambda = \mathcal{C}_0[\Delta_V] = \frac{1}{64}\delta_0$ and $\Xi_\lambda = \Delta_V - \iota_0[\Phi_\lambda] = 0$ by Lemma 2.3(b). The fine charge is positive: $\|\Delta_V\|^2 = \|\frac{1}{64}\delta_0\|^2 = \frac{9}{4096}$, so $\|\Delta_V\| = \frac{3}{64}$, and, the family being process-induced, by the three-point bound of Corollary 2.8 with $\min p_2 = \frac{1}{3}$, $\frac{3}{64\sqrt{5}} \leq h_3(V) \leq \frac{3}{64}$.

Production. With $\Pi_1 = R_1 L$ for the uniform law, $Q_1 = I - \Pi_1$ averages the first two coordinates against each other: $Q_1 A$ has rows $\frac{1}{2}(\mathrm{row}_1 - \mathrm{row}_2)(A) = (1/8,\, 0,\, -1/8)$, its negative, and zero. Then

$$B\,Q_1 A = \begin{pmatrix} 0 & 0 & 0 \\ -1/32 & 0 & 1/32 \\ 1/32 & 0 & -1/32 \end{pmatrix}, \qquad \mathrm{P}_\lambda(2,0,1) = L\,B\,Q_1 A\,R_0 = \begin{pmatrix} -1/64 & 1/32 \\ 1/64 & -1/32 \end{pmatrix} = -\tfrac{1}{64}\,\delta_0,$$

which is nonzero. Thus $\mathrm{P}_\lambda = -\Phi_\lambda$, the balance law gives $\Delta_{V^\lambda} = \Phi_\lambda + \mathrm{P}_\lambda = 0$, and one checks directly that $LV(2 \leftarrow 0)R_0 = \big(LBR_1\big)\big(LAR_0\big)$. The cross term is $\langle \Phi_\lambda, \mathrm{P}_\lambda \rangle = -\|\frac{1}{64}\delta_0\|^2 = -\frac{9}{4096}$. Every equality above is an identity of rational matrices. □

Remark 6.6 (The interference dial)*.* The construction is a one-parameter family: replace $\frac{1}{64}$ by x in $V(2 \leftarrow 0) = BA + x\,\iota_0[\delta_0]$; the kernel is entrywise non-negative for $x \in [-\frac{5}{32}, \frac{5}{16}]$ and the joint law π is non-negative for $x \in [-\frac{3}{32}, \frac{3}{16}]$. Production depends only on A, B, and the cut (Proposition 4.9), so $\mathrm{P}_\lambda = -\frac{1}{64}\delta_0$ for every x, while $\Phi_\lambda = x\,\delta_0$; the balance law gives

$$\|\Delta_V\| \;=\; 3\,|x|, \qquad \|\Delta_{V^\lambda}\| \;=\; 3\,\big|x - \tfrac{1}{64}\big|.$$

The coarse deficit is the transmitted flux $x\,\delta_0$ less the fixed production $\frac{1}{64}\delta_0$. Within this family, in which the production term and the flux are collinear, the parameter moves the cut through coherent production ($x < 0$), pure production from a flat fine family ($x = 0$), exact conservation of the deficit norm ($x = \frac{1}{128}$), cancellation ($x = \frac{1}{64}$), and growth with reversed orientation ($x > \frac{1}{64}$). The general balance law allows arbitrary relative orientations of flux and production and nonzero severing; the collinear family is the one-dimensional case of Proposition 4.9.

Remark 6.7. Proposition 6.5 also shows that for charged fine families, flatness of the observed family does not certify a closed cut: here V^λ is flat while $\mathrm{P}_\lambda \neq 0$. The equivalence of Corollary 4.6 between vanishing production and coarse flatness uses flatness of the fine family; for a charged fine family only the balance law holds.

Remark 6.8 (The lumped process is not Markov). The joint law π of Proposition 6.5 determines the law of the lumped process $Y = \lambda(X)$ on $\{a, b\}^3$, and its three-time conditionals do not reduce to two-time conditionals:

$$\mathbb{P}(Y_2 = a \mid Y_1 = a,\, Y_0 = a) = \tfrac{37}{60}, \qquad \mathbb{P}(Y_2 = a \mid Y_1 = a,\, Y_0 = b) = \tfrac{23}{36}, \qquad \mathbb{P}(Y_2 = a \mid Y_1 = a) = \tfrac{5}{8},$$

and the largest difference $\big|\mathbb{P}(Y_2 \mid Y_1, Y_0) - \mathbb{P}(Y_2 \mid Y_1)\big|$ over the eight conditionings is $\frac{1}{24}$. So the two-time family of Y composes while Y depends on its past beyond one step. The interference cut cancels the two-time composition defect, the quantity the balance law tracks, and leaves the higher-order dependence of the lumped process in place. Every occurrence of "memory" in this paper refers to the two-time composition defect.

7 Exact examples

Every quantity in this paper is computed in exact arithmetic and is a rational number or an algebraic number in closed form; there is no numerical optimization, no simulation, and no data, and the decimal approximations printed for orientation are outward-rounded and enter no argument. Positivity of a charge is certified by exact positivity of the deficit norm, since $\mathcal{F}\mathrm{lat}_n$ is the zero set of the deficit; flatness is certified by exact vanishing; and the charges of the charged families are given as the exact windows of Corollary 2.8, all four families being process-induced. The metric is (6) with the families' own laws and unit pair and triple weights. Table 1 lists the four families worked in the text. Each row satisfies the balance identity $\|\Delta_{V^\lambda}\|^2 = \|\Phi_\lambda\|^2 + 2\langle\Phi_\lambda, \mathrm{P}_\lambda\rangle + \|\mathrm{P}_\lambda\|^2$ and the conservation identity $\|\Delta_V\|^2 = \|\Phi_\lambda\|^2 + \|\Xi_\lambda\|^2$ as identities of rational numbers. The rows are illustrations and consistency checks of Theorem 4.2 on four families; the theorem itself is proved in Section 4. The four rows are extremal mechanisms of the balance law: pure production, pure severing, closure, and perfect destructive interference. The general cut has nonzero flux, severed part, and production in arbitrary relative orientation, and does not fall into one of the four classes.

The first row of Table 1 creates deficit from a flat family through production alone; the third row is the same cut geometry made dynamically closed, and every entry vanishes. The second row destroys the deficit with zero production, through severing, and its fine entry factorizes as the law-metric size of the flat companion kernel times the discarded factor's deficit, $153/188600 = (17/16) \cdot (18/23575)$. The fourth row is the only one with a nonzero cross term, and the cross term is extremal, $\langle\Phi_\lambda, \mathrm{P}_\lambda\rangle = -\|\Phi_\lambda\|\|\mathrm{P}_\lambda\|$. The dial of Remark 6.6 moves a single family through the first and fourth rows and the collinear cases between them, with $\|\Delta_V\| = 3|x|$ and $\|\Delta_{V^\lambda}\| = 3\,|x - \frac{1}{64}|$.

Table 1: Four extremal mechanisms of the balance law. All entries are exact; squared norms and inner products are taken in the metric (6) at the appropriate level, deficit cochains summed over triples.

Family	$\|\Delta_V\|^2$	$\|\Phi_\lambda\|^2$	$\|\Xi_\lambda\|^2$	$\|\mathrm{P}_\lambda\|^2$	$\langle\Phi_\lambda, \mathrm{P}_\lambda\rangle$	$\|\Delta_{V^\lambda}\|^2$
Ex. 3.2 (production)	0	0	0	18/23575	0	18/23575
Ex. 3.3 (severing)	153/188600	0	153/188600	0	0	0
Ex. 3.4 (closed cut)	0	0	0	0	0	0
Prop. 6.5 (interference)	9/4096	9/4096	0	9/4096	−9/4096	0

Table 2: Exact windows for the charges of the families of Table 1, from the three-point bound of Corollary 2.8 with each family's own laws. A flat family has charge 0.

Family	$h_N(V)$ (fine)	$h_n(V^\lambda)$ (coarse)
Ex. 3.2	0	$\sqrt{9/56375} \le h_2 \le \sqrt{18/23575}$
Ex. 3.3	$\sqrt{51/475600} \le h_4 \le \sqrt{153/188600}$	0
Ex. 3.4	0	0
Prop. 6.5	$3/(64\sqrt{5}) \le h_3 \le 3/64$	0

8 Conclusion

The exact object of this paper is the deficit-level balance $\Delta_{V^\lambda} = \Phi_\lambda + \mathrm{P}_\lambda$ together with the orthogonal splitting $\|\Delta_V\|^2 = \|\Phi_\lambda\|^2 + \|\Xi_\lambda\|^2$. Coarse-graining changes the two-time composition defect through three mechanisms: transmission, which is contractive; severing, which removes the part of the deficit orthogonal to the block-measurable subspace; and production, which is sourced at the cut and can add to or cancel the transmitted part. The dimension-matched charge inherits quantitative information from the deficit through the two-sided comparison of Proposition 2.6, with explicit constants and exponent one, and not through an additive law of its own.

The balance law concerns the two-time deficit: two-time flatness of the lumped family does not imply that the lumped process is Markov (Corollary 4.6, Remark 6.8), and the severed and produced quantities say nothing about higher-order dependence. The identification of production with a Mori–Zwanzig memory kernel and the equality of that kernel with the observed deficit hold for flat fine families (Theorem 5.2); for a charged fine family the production term differs from the coarse deficit by the flux, and Remark 6.7 gives a flat coarse family with nonzero production. The uniqueness statement of Theorem 5.2(iii) is among block-supported stochastic sections. The nearest flat family, and with it the residual, is not unique in general (Example 5.4, a family with laws that is not process-induced); the charge is unique. The orthogonality, the contraction of the flux, and the sizes of the terms are properties of the law-weighted metric (6), while the identities between kernels and the zero sets hold in every metric. The theory concerns the dimension-matched charge h_n and not the full liftability profile; at $d = n$ the charge and the deficit norm are comparable (Proposition 2.6), and the information in the profile beyond the deficit norm lies at $d > n$ (Remark 2.7).

9 Open problems

(i) *Coherent cuts.* By Proposition 4.9, for fixed short kernels the production term is fixed and the flux is affine in the long kernel, so hypothesis (C) at a triple is a half-space condition on the compression of the long kernel. The long kernel of a process-induced family is constrained by the existence of a joint law with the given two-time margins; the open question is when that constraint places the flux in the half-space. In the collinear family of Remark 6.6 the answer is the sign of x; in general the constraint set is the image of a marginal problem with three prescribed two-time margins.

(ii) *Balance for the full profile.* This paper works at the dimension-matched level h_n. For $d > n$ the model sets $\mathcal{F}_d$ of [22] are closures of compressions of d-state flat transports, strictly larger than $\mathcal{F}\mathrm{lat}_n$, and are not graphs over one-step kernels, so the argument of Proposition 2.6 does not apply. A balance law for h_d would have production sourced only by the part of the cut not explained by $d - n$ hidden states, and the memory rank of the lumped family against the rank of the fine one is the discrete form of the question. The problem is to formulate and prove such a law.

(iii) *Continuous grids.* The first question on refining grids is the scaling of the deficit and of the production term with the time increment: the two can scale at different orders depending on the short-time structure of the family, and a production rate exists only if a normalization makes the production over a window converge. Under such a normalization the balance law would become a continuity equation for the two-time composition defect along the coarse-graining semigroup, with cuts closed on the accessible subspace as the vanishing-rate case. The relation to entropy production along a cut in stochastic thermodynamics is an analogy at this stage. A second question, raised by Remark 6.8, is what the severed term captures of the higher-order dependence of the lumped process, if anything.

(iv) *Estimation and a test of two-time closure.* With the laws held fixed, flux and production are Lipschitz in the family by construction, so plug-in estimates inherit the convergence of the family estimator as in [22, Proposition 4.6]. In an application the laws are estimated too, and they enter through the disintegrations R_i, whose entries divide by the block masses; a usable statement needs a margin $\min_{i,a} p_i^{\lambda}(a) \geq \eta > 0$ and joint propagation of the errors in the family and in the laws. A test of $\mathrm{P}_\lambda = 0$ from an estimated family is a test of closure on the accessible subspace, hence of two-time Chapman–Kolmogorov lumpability; it is not a test of Markovianity of the lumped process, which needs higher-order conditionals. The bounds of Corollary 4.8 relate the production statistic to the charge and are not a test; a test needs the null distribution of the statistic when the kernels and the laws are both estimated, which is open. The empirical counterpart is the Chapman–Kolmogorov test of Markov-state-model validation [23, 13], which tests an estimated coarse family for composition at multiples of the lag time and assesses the outcome by resampling.

(v) *Optimal cuts.* Among cuts of fixed coarse dimension, minimizing production is a model-reduction objective related to the optimal-prediction program [3, 9]: it seeks the coarse description closest to dynamical autonomy rather than the most predictive one. For a flat fine family, minimizing production and minimizing the charge of the observed family have the same zero set and are two-sidedly comparable by Corollary 4.8; they need not order the cuts identically, since the constants depend on the coarse laws, which depend on the cut. The size of the gap between production-minimizing and prediction-optimal cuts, and the relation of production-minimizing cuts to the spectral structure of the fine generator when one exists, are open.

(vi) *The upper constant.* On the three-point grid $h_n(V) \leq \|\Delta_V\|$ (Proposition 2.6). On a grid of $m+1$ points the argument gives $C_*(m, \rho)$, and $(m(m-1)/2)^{1/2}$ for families that transport their laws (Corollary 2.8), because the flat family generated by the one-step kernels of V is compared with the deficits of the consecutive-cut triples only. The problem is whether $h_n(V) \leq \|\Delta_V\|$ holds

on every grid, and, if not, what the sharp constant is.

(vii) *Uniqueness of the nearest flat family for process-induced families.* Example 5.4 has two nearest flat families and is not process-induced. The problem is whether a process-induced family can have more than one nearest flat family. A negative answer would make the residual of Remark 5.3 canonical on the process-induced families, the class the theory concerns.

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